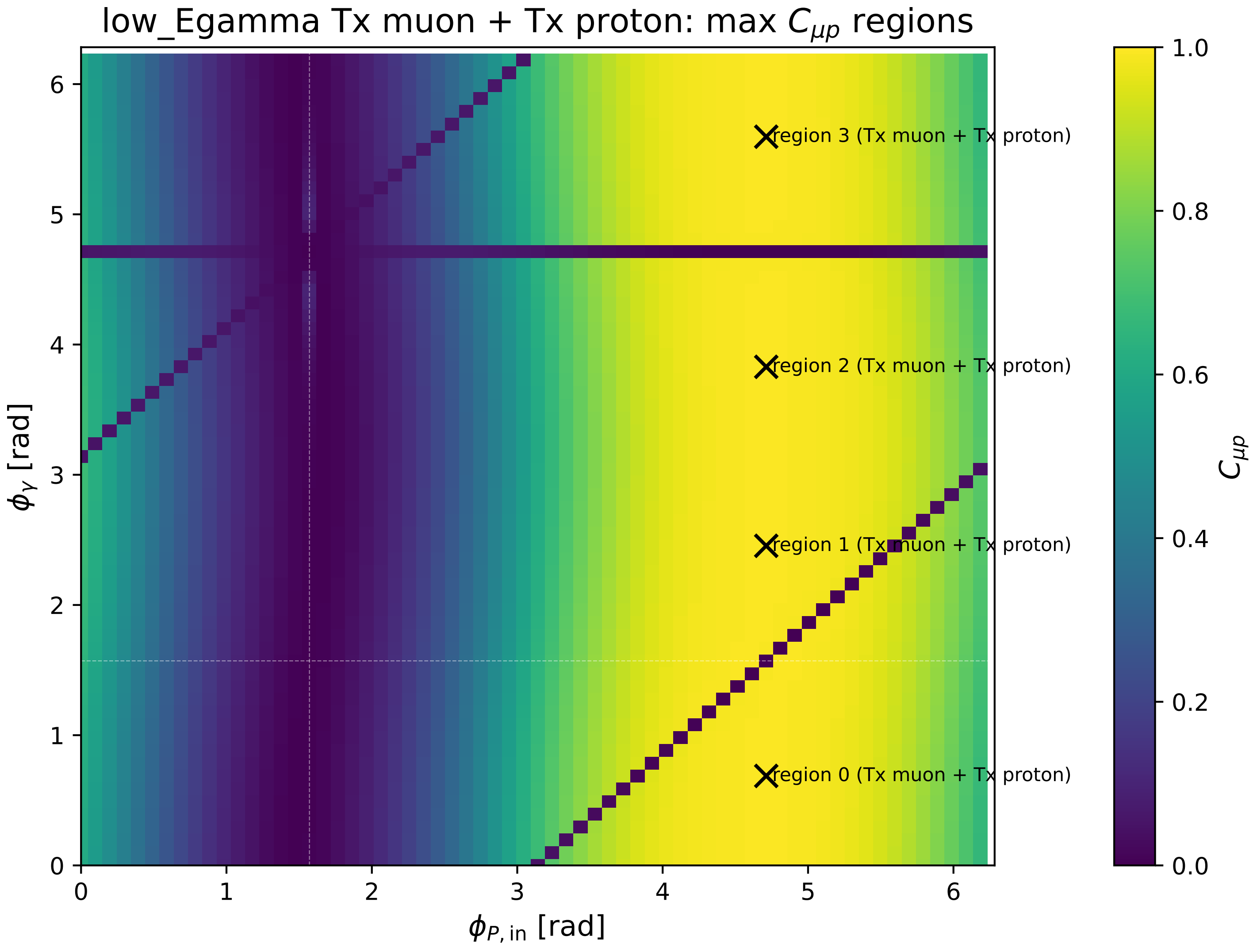
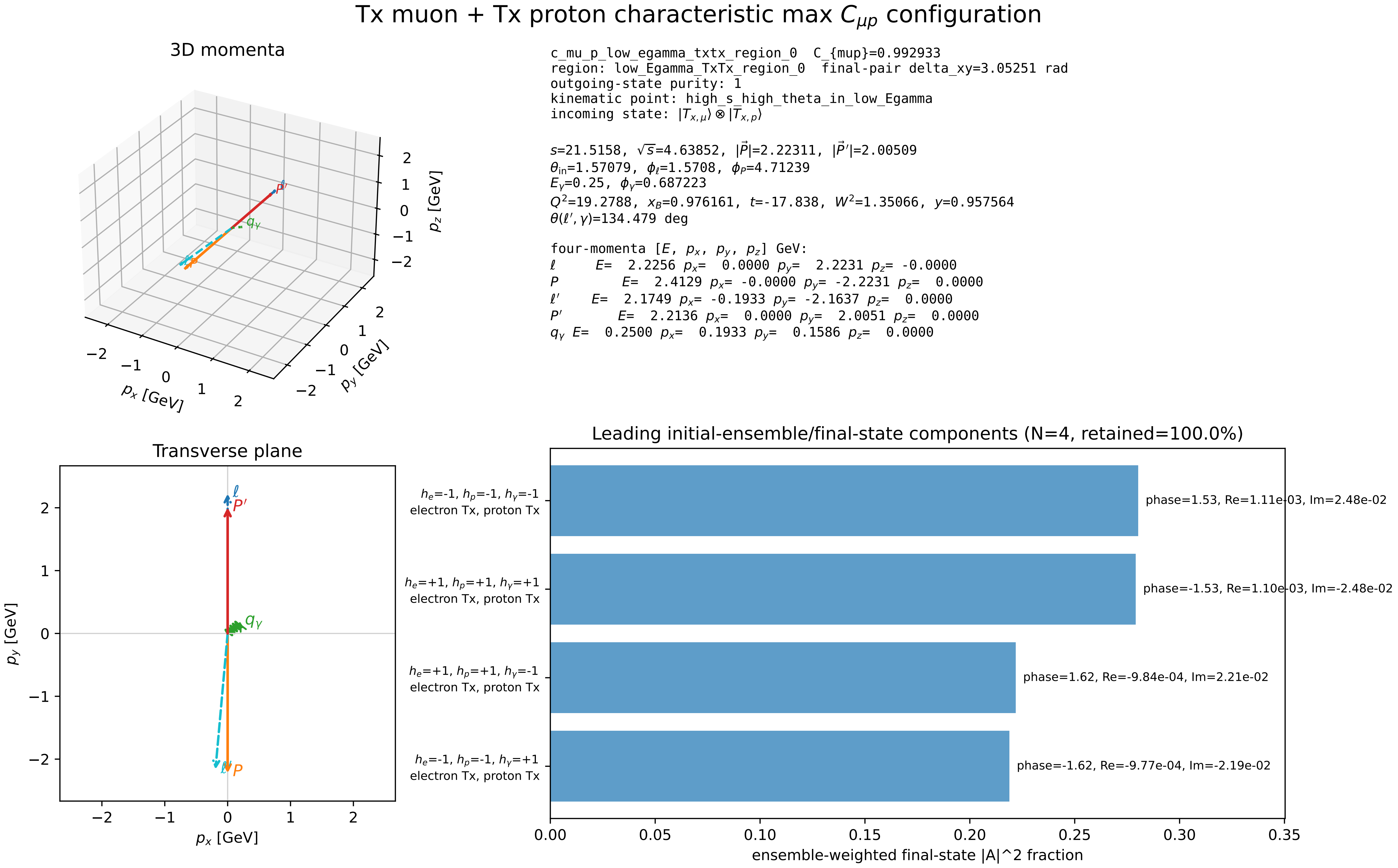
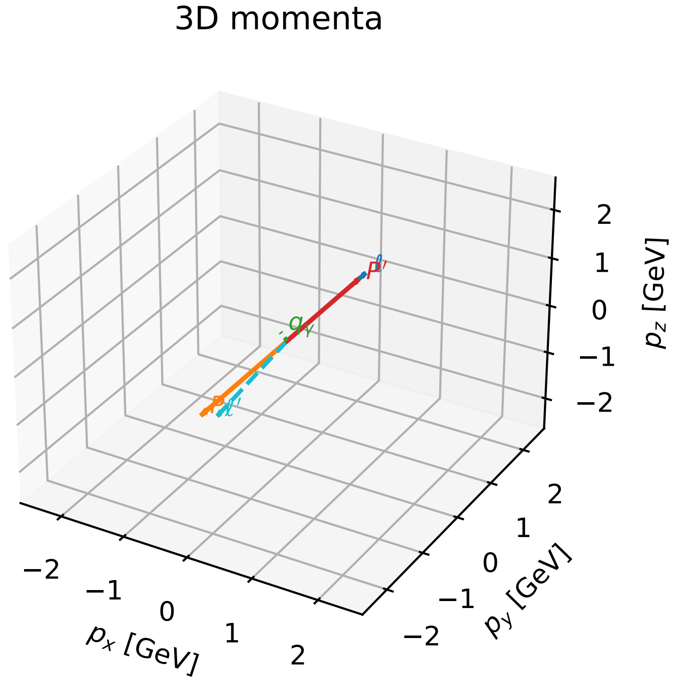

low_Egamma Tx muon + Tx proton: max $C_{\mu p}$ regions
region 3 (Tx muon + Tx proton)
region 2 (Tx muon + Tx proton)
region 1 (Tx muon + Tx proton)
region 0 (Tx muon + Tx proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\mu p}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0


Tx muon + Tx proton characteristic max $C_{\mu p}$ configuration
3D momenta
c_mu_p_low_egamma_txtx_region_0 C_{mup}=0.992933
region: low_Egamma_TxTx_region_0 final-pair delta_xy=3.05251 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.00509
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=0.687223
Q^2=19.2788, x_B=0.976161, t=-17.838, W^2=1.35066, y=0.957564
$\theta(\ell',\gamma)$=134.479 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 0.0000 p_y= 2.2231 p_z= -0.0000
P E= 2.4129 p_x= -0.0000 p_y= -2.2231 p_z= 0.0000
ℓ' E= 2.1749 p_x= -0.1933 p_y= -2.1637 p_z= 0.0000
P' E= 2.2136 p_x= 0.0000 p_y= 2.0051 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1933 p_y= 0.1586 p_z= 0.0000
p_x [GeV]
p_y [GeV]
p_z [GeV]
Transverse plane
ℓ
P'
q_γ
ℓ'
P
Leading initial-ensemble/final-state components (N=4, retained=100.0%)
h_e=-1, h_p=-1, h_γ=-1 electron Tx, proton Tx
phase=1.53, Re=1.11e-03, Im=2.48e-02
h_e=+1, h_p=+1, h_γ=+1 electron Tx, proton Tx
phase=-1.53, Re=1.10e-03, Im=-2.48e-02
h_e=+1, h_p=+1, h_γ=-1 electron Tx, proton Tx
phase=1.62, Re=-9.84e-04, Im=2.21e-02
h_e=-1, h_p=-1, h_γ=+1 electron Tx, proton Tx
phase=-1.62, Re=-9.77e-04, Im=-2.19e-02
ensemble-weighted final-state |A|^2 fraction

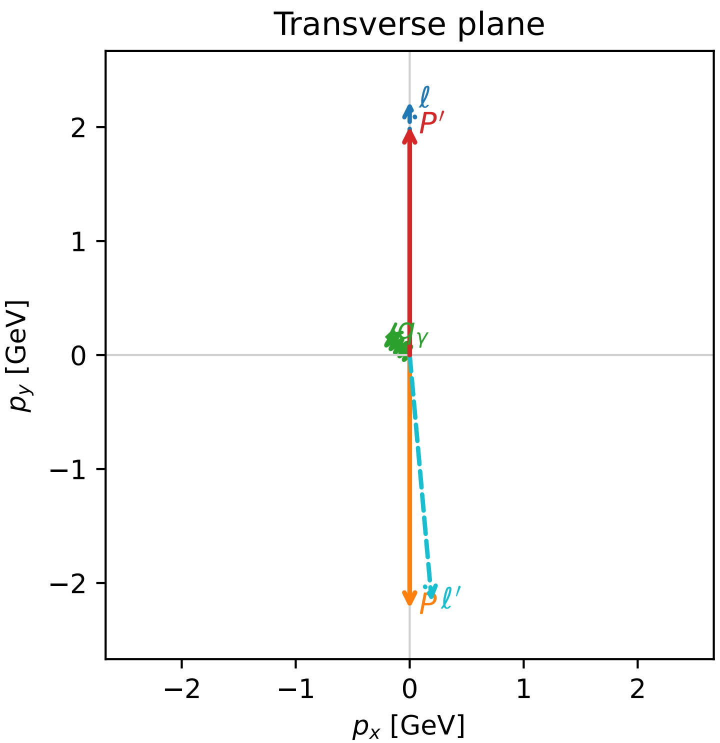
Tx muon + Tx proton characteristic max $C_{\mu p}$ configuration

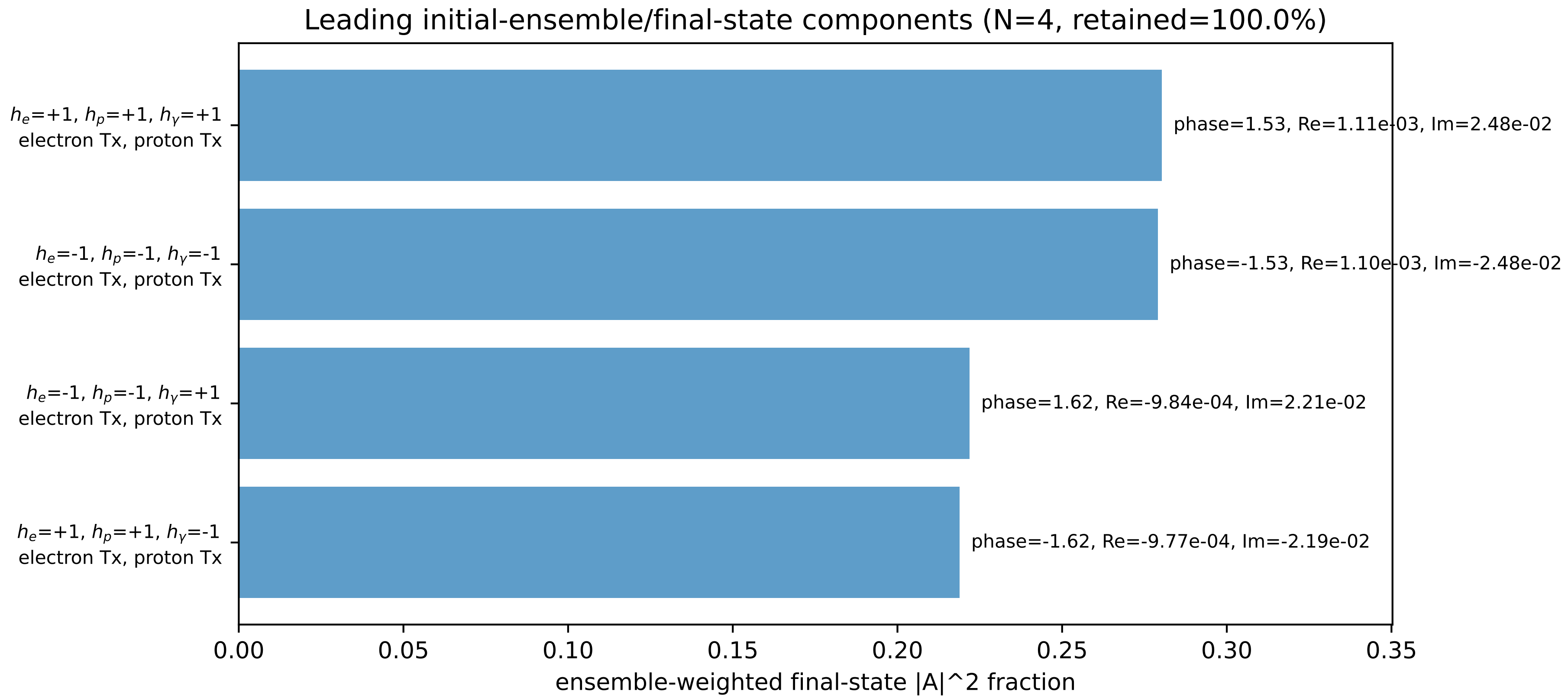



```
c_mu_p_low_egamma_txtx_region_1  C_{mup}=0.992933
region: low_Egamma_TxTx_region_1  final-pair delta_xy=3.05251 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
```
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

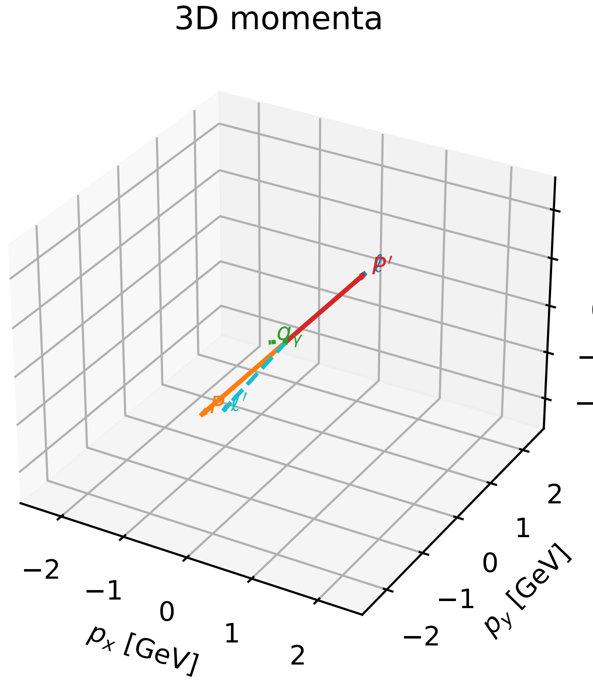
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.00509
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=2.45437
Q^2=19.2788, x_B=0.976161, t=-17.838, W^2=1.35066, y=0.957564
$\theta(\ell', \gamma)$=134.479 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 0.0000 p_y= 2.2231 p_z= -0.0000
P E= 2.4129 p_x= -0.0000 p_y= -2.2231 p_z= 0.0000
ℓ' E= 2.1749 p_x= 0.1933 p_y= -2.1637 p_z= 0.0000
P' E= 2.2136 p_x= 0.0000 p_y= 2.0051 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1933 p_y= 0.1586 p_z= 0.0000





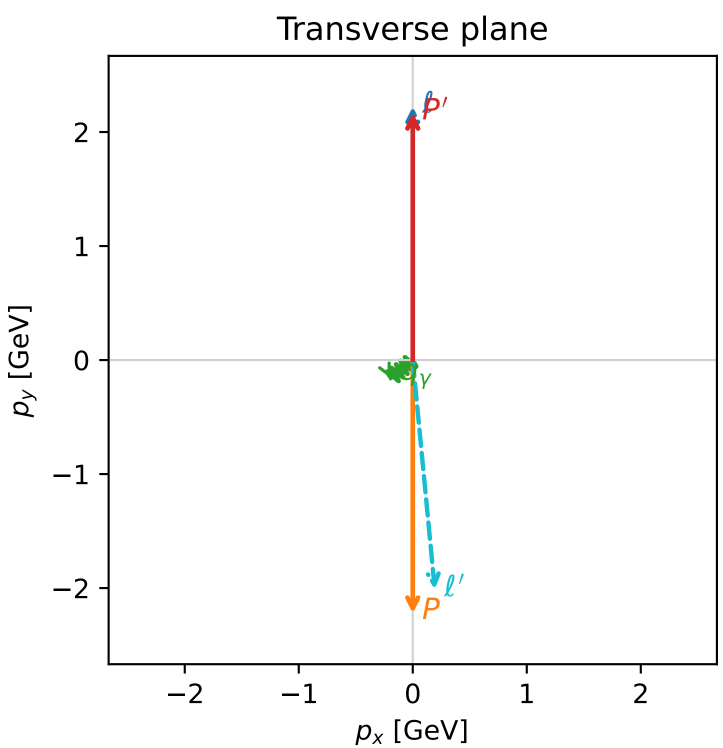
Tx muon + Tx proton characteristic max $C_{\mu p}$ configuration

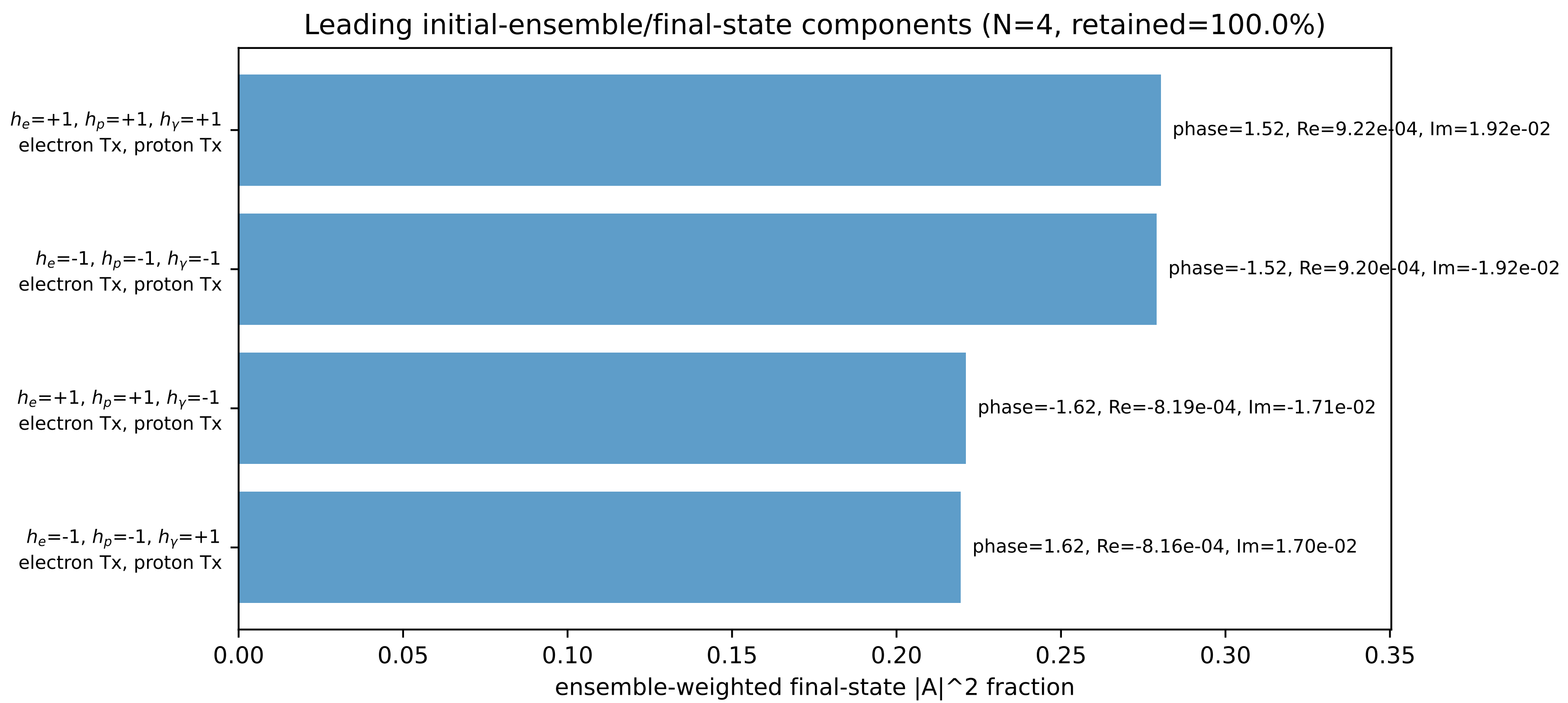


c_mu_p_low_egamma_txtx_region_2 C_{mup}=0.992908
region: low_Egamma_TxTx_region_2 final-pair delta_xy=3.04583 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

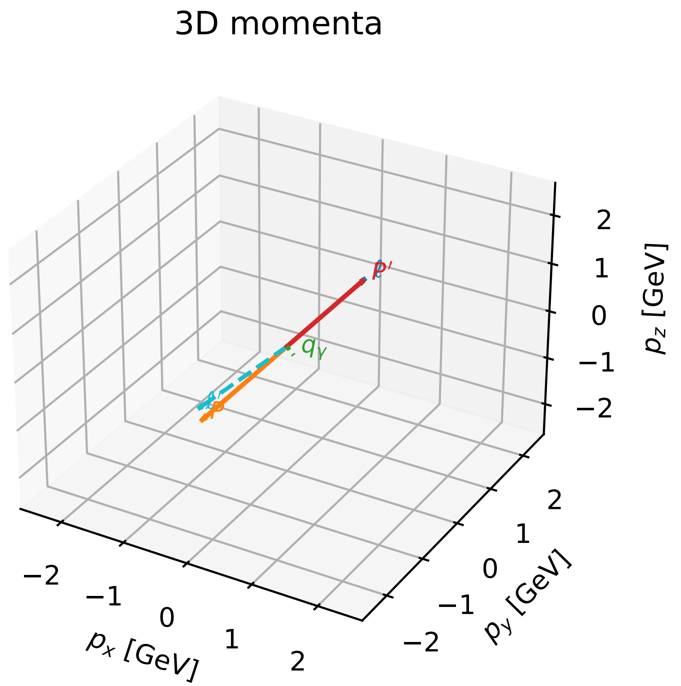
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.17054
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=3.82882
Q^2=17.9324, x_B=0.905532, t=-19.3019, W^2=2.75061, y=0.960161
$\theta(\ell', \gamma)$=56.1116 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= 0.0000 p_y= 2.2231 p_z= -0.0000
P E= 2.4129 p_x= -0.0000 p_y= -2.2231 p_z= 0.0000
ℓ' E= 2.0240 p_x= 0.1933 p_y= -2.0119 p_z= 0.0000
P' E= 2.3646 p_x= 0.0000 p_y= 2.1705 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1933 p_y= -0.1586 p_z= 0.0000





Tx muon + Tx proton characteristic max $C_{\mu p}$ configuration



c_mu_p_low_egamma_txtx_region_3 C_{mup}=0.992908
region: low_Egamma_TxTx_region_3 final-pair delta_xy=3.04583 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|T_{x,\mu}\rangle \otimes |T_{x,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.17054
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=5.59596
Q^2=17.9324, x_B=0.905532, t=-19.3019, W^2=2.75061, y=0.960161
$\theta(\ell', \gamma)$=56.1116 deg

four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.2256	p_x= 0.0000	p_y= 2.2231	p_z= -0.0000
P	E= 2.4129	p_x= -0.0000	p_y= -2.2231	p_z= 0.0000
ℓ'	E= 2.0240	p_x= -0.1933	p_y= -2.0119	p_z= 0.0000
P'	E= 2.3646	p_x= 0.0000	p_y= 2.1705	p_z= 0.0000
q_γ	E= 0.2500	p_x= 0.1933	p_y= -0.1586	p_z= 0.0000